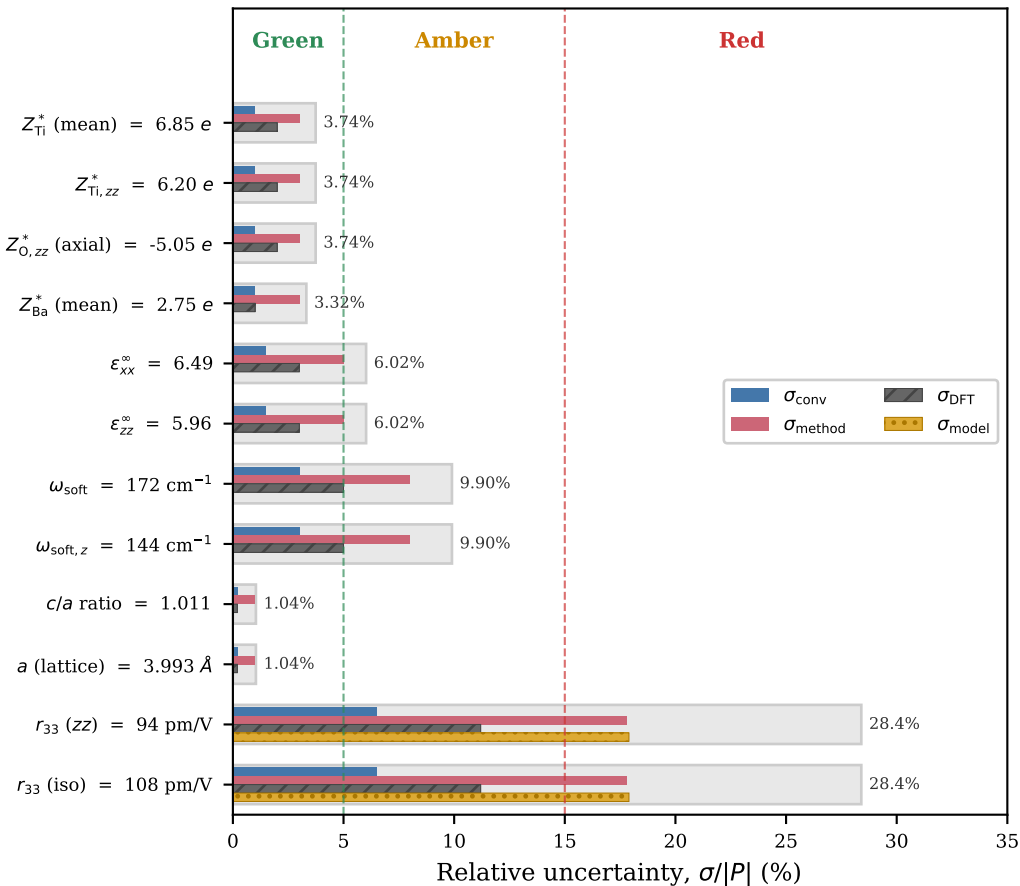


**a) Certificate uncertainty decomposition**



**b) Example JSON certificate**

```
{
  "parameter": "r_33_iso",
  "value": 108.3,
  "unit": "pm/V",
  "uncertainties": {
    "convergence": 6.5,
    "methodological": 17.8,
    "intrinsic_dft": 11.2,
    "model_form": 17.9,
    "total": 28.4
  },
  "quality_tier": "Red",
  "provenance": {
    "code": "Quantum ESPRESSO",
    "version": "7.5",
    "functional": "PBE",
    "pseudopotential": "PAW",
    "phase": "tetragonal",
    "tensor_component": "all-isotropic",
    "energy_cutoff_Ry": 60,
    "kpoint_mesh": [
      6,
      6,
      6
    ],
    "supercell": "2x2x2",
    "n_atoms": 40
  },
  "calibration": {
    "Z_ref_e": 7.16,
    "epsilon_ref": 6.5,
    "model": "soft-mode r33"
  },
  "schema_version": "posterior-certificate-v1"
}
```

All uncertainty components are relative percentages.  
The full table contains one record per certified property.